GOLDEN GATE UNIVERSITY

Doctor of Business Administration (Generative AI)

DBA Topic Proposal v2.1

Enhanced Structure Following GGU + High-Impact DBA Skeleton

A Governance Framework for Responsible Explainable AI-Assisted Retrospective EEG-Based Neuropsychiatric Decision Support Under Human Clinical Oversight

Integrating Generative AI, Retrieval-Augmented Generation, and Continuous Monitoring for Clinical and Consumer Healthcare Applications

Author: Praveen K Asthana

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Strictly follows the enhanced DBA dissertation skeleton:
Front Matter → Abstract → Ch.1–5 → Appendices A–J.
Every heading and every table carries a mandatory 2-line introduction.
Presentation: tabularx + booktabs + TikZ flowcharts throughout.

Declaration

This topic proposal is a planning artefact for the candidate's DBA dissertation; it states what the dissertation will deliver, not the dissertation itself. The full empirical study, governance framework, and validation evidence are deferred to the main thesis under separate IRB scope.

Declaration item	Statement
Originality	The proposal and underlying research are the candidate's own work; no fabrication or plagiarism.
AI tool disclosure	ChatGPT/Codex and Claude were used for editorial assistance and structuring; all decisions, evidence, and citations are verified by the candidate (see Appendix H).
Prior IRB approval	Streams 3 + 4 (n=131 PLS-SEM, n=12 expert panel) are cited as aggregated, anonymised, prior-IRB-approved secondary data only (references [8]–[10] included; prior IRB certificate numbers still to be inserted before submission).
Submission readiness	Topic proposal → Main dissertation alignment verified via separate audit (`ALIGNMENT_AUDIT_proposal_vs_main_thesis.md`).

Certificate / Supervisor Approval

This certificate carries the supervisor's confirmation that the topic, scope, and methodology described herein are appropriate for a Doctor of Business Administration submission. Signature blocks are left blank for institutional signing and remain a release blocker before final submission.

Role	Name (to insert)	Signature / Date
Doctoral Supervisor	[supervisor name + title]	________________
Co-Supervisor / Chair	[chair name + title]	________________
DBA Programme Director	[director name]	________________

Acknowledgements

The candidate acknowledges the doctoral supervisor, the cohort peer-review group, and the data custodians of the publicly available KAU and ABIDE-II datasets. Acknowledgement of the prior-IRB-approved n=131 clinician and n=12 expert-panel data sources is recorded separately in Appendix H (AI Report Approval Form) and the AI Declaration.

Contents

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List of Abbreviations

The proposal uses domain abbreviations consistently with the main dissertation; the list below is the canonical reference for examiners. Any abbreviation introduced in only one chapter is defined on first use in that chapter and is not repeated here.

Abbrev.	Expansion
ASD	Autism Spectrum Disorder
EEG	Electroencephalogram
RGAIG	Responsible Generative AI Governance (proposed framework)
RAG	Retrieval-Augmented Generation
HITL	Human-In-The-Loop
XAI	Explainable AI
RAI	Responsible AI
DSR	Design Science Research
PLS-SEM	Partial Least Squares Structural Equation Modelling
TAM / TOE / RBV	Technology Acceptance / Technology-Organisation-Environment / Resource-Based View
KAU / ABIDE-II	Saudi reference EEG dataset / Western multi-site ASD dataset
NIST AI RMF	US National AI Risk Management Framework
ISO/IEC 42001	International AI Management System standard
EU AI Act	European Union Artificial Intelligence Act
HIPAA / GDPR / DPDP	US health-data / EU general data / India data-protection laws

Glossary of Terms

A short glossary disambiguates the multi-disciplinary vocabulary spanning AI engineering, governance, healthcare, and DBA practice. Each entry is the operational definition used throughout this proposal and the planned dissertation.

Term	Operational definition
Agentic AI	A multi-step AI workflow with planner, decomposer, policy engine, and human-in-the-loop — each step independently auditable.
Responsible AI	The combined practice of fairness, accountability, transparency, robustness, privacy, and contestability throughout the AI lifecycle.
Governance maturity	A measurable index of how reliably an organisation governs AI risk, ranging from ad-hoc to optimising.
B2C-primary / B2B-supporting	A deployment posture where the caregiver / family is the primary beneficiary, and the clinician / hospital provides the safety, escalation, and governance layer.
Decision-audit row	A persisted record of one AI decision containing request ID, inputs, prompt version, model version, output, confidence, rules applied, and human override.

1. Abstract

The abstract distils the proposal into seven structured movements covering background, problem, aim, method, findings expectation, contribution, and keywords. Examiners read the abstract first; the table below mirrors the seven-part skeleton mandated by the enhanced DBA structure.

Abstract movement	Content of this proposal
1. Research Background	ASD prevalence rising [1], [2]; pediatric screening burden in tier-1/tier-2 institutions; under-representation of Indian cohorts; growing Generative AI capability but immature governance fabric.
2. Research Problem	Enterprise / clinical AI adoption is failing trust, explainability, accountability, and adoption gates simultaneously; current frameworks are either technical (NIST, ISO) [11], [12] or theoretical (TAM, TOE) [5], [6] but rarely both.
3. Aim and Objectives	Develop & evaluate a Responsible Generative AI Governance (RGAIG) framework for B2C-primary, clinician-supported ASD screening support; 6 objectives, 8 RQs, 5 hypotheses.
4. Methodology	Pragmatist mixed-methods Design Science Research [8], [9] + PLS-SEM [10] + 12-expert qualitative panel + retrospective benchmarks on KAU [3] + ABIDE-II [4] datasets.
5. Key Findings (expected)	H1–H5 evidence that governance maturity mediates between RAI principles and adoption outcomes; explainability + emotional reassurance dominate caregiver trust.
6. Contribution	Theoretical (RGAIG construct + 525-module taxonomy), practical (maturity instrument), methodological (DSR + PLS-SEM hybrid), policy (multi-jurisdiction governance mapping).
7. Keywords	Generative AI; Responsible AI; AI Governance; Enterprise AI; Agentic AI; Explainable AI; Digital Transformation; ASD; EEG; Healthcare AI.

2. GGU Topic Proposal Requirement Map

This section converts the university's administrative Topic Proposal requirements into an explicit compliance map. It is placed immediately after the abstract so the Chair and DDDR can verify that the proposal contains each required item before reading the detailed DBA evidence tables.

GGU-required item	Where this proposal addresses it	Readiness
Topic and examination context	Title page; Table T1; Topic Context & Problem Decomposition; B2C-primary / B2B-supporting framing for pediatric ASD screening support.	Complete
Importance / expected contribution	Table T1; Expected Solution Outcomes; Value-Creation Table; Examiner Most-Looked-For Items; contribution row in Abstract.	Complete
Review of relevant literature	Chapter 2 skeleton; Theme-Based Clustering; Comparative Matrix; Research Gap typology; Research Framework figure.	Complete for topic stage
Continuity with prior research	Literature themes connect NIST AI RMF [11], ISO/IEC 42001 [12], TAM [5] / TOE [6] / RBV [7], RAI [20], XAI [17], [18], RAG [19], HITL, and ASD screening literature [1]–[4] to the RGAIG contribution.	Complete for topic stage
Expected research design	Chapter 3 skeleton; Design Science Research process; Sampling Strategy; Ethics; Risk Register; Timeline.	Complete
Preliminary statistical tools	Hypotheses H1–H5; PLS-SEM; reliability / validity planning; findings traceability matrix; planned quantitative analysis in Chapter 4.	Complete
References	References are now included as a numbered section with bracketed citations across key claims.	Complete; final style audit recommended
Committee approval pathway	Certificate / Supervisor Approval; Submission Checklist; release blockers identified for supervisor, chair, director, and prior-IRB details.	Pending signatures

Table 1: GGU topic-proposal requirement map — administrative requirement to proposal evidence.

3. Topic Context & Problem Decomposition

This section delivers the seven operator-requested content blocks that anchor every later chapter: topic + why this topic, problem + sub-problems, 5-level pain articulation, B2C side, B2B side, time-to-onboard analysis, and AS IS vs TO BE process comparison. The tables below stand on their own so an examiner can read this single section and grasp the entire problem space before drilling into the chapter skeletons.

3.1 Table T1 — Topic Statement & Why This Topic (10 Reasons)

The topic is fixed in one sentence and then defended with ten orthogonal reasons spanning academic, business, societal, governance, and feasibility dimensions. A topic that cannot answer "why this and not something else" is not yet a DBA topic; each row below is one defendable answer to that examiner question.

#	Reason category	Why this topic (one-line defence)
Topic. Governance-enabled, B2C-primary, B2B-supporting AI screening support for ASD (RGAIG framework).		
1	Academic novelty	No prior work integrates RAI [20] + agentic-RAG [19] + caregiver-trust + clinician-HITL into one auditable governance fabric for pediatric ASD.
2	Practitioner urgency	3–12 month specialist wait; family carries the burden between assessments; AI support is non-diagnostic but reduces uncertainty.
3	Industry adoption gap	Healthcare-AI adoption fails on trust, explainability, and governance more than on accuracy.
4	Policy timing	EU AI Act Art. 86 [13] + ISO/IEC 42001 [12] + NIST AI RMF [11] all live; a governance artefact aligned to all three is timely and useful.
5	Methodological fit	DSR [8], [9] + PLS-SEM [10] + qualitative panel is a high-evidence design for a governance artefact.
6	Societal impact	ASD prevalence rising [1], [2]; Indian pediatric cohort under-represented; sex-skew $\sim$ 4:1 ignored in most datasets.
7	Feasibility	Candidate's `agenticfinder` reference codebase (305 GB; 198 trained models) supplies the technical anchor; no greenfield engineering required.
8	Ethical safeguards	Pediatric / vulnerable-population pathway forces explicit safeguards that strengthen the contribution.
9	Defensibility	Single-disease focus (ASD) is auditable; scope creep into multi-disease is explicitly excluded.
10	Originality	RGAIG construct + 525-module quality taxonomy + 5-layer enterprise framework are new artefacts, not re-packaged literature.

Table 2: Topic statement + 10 reasons (sufficient justification).

3.2 Table T2 — Problem & Sub-Problems (Decomposition)

The headline problem is decomposed into eight orthogonal sub-problems that the dissertation must each address with evidence, design, or policy. A clear sub-problem decomposition is the antidote to "vague DBA" — every later chapter maps back to one numbered sub-problem.

#	Sub-problem	What it manifests as in practice
Headline problem. Pediatric ASD screening support is fragmented, unreliable, and untrustworthy at the caregiver–clinician interface.		
SP1	Diagnostic-wait limbo	3–12 month specialist wait; family in emotional uncertainty with no decision support.
SP2	Unreadable reports	ADOS / DSM-5 clinical jargon; caregivers cannot translate findings into action.
SP3	Black-box AI distrust	Caregivers reject AI outputs without explanation; clinicians reject without provenance.
SP4	Governance gap	No unified mapping across NIST [11] + ISO [12] + EU AI Act [13] + DPDP [16] for pediatric AI.
SP5	Adoption gap	Even excellent models stall at hospital procurement on missing audit + safety + role boundaries.
SP6	Caregiver overload	No structured peer / professional support loop tied to the screening process.
SP7	Indian-cohort under-representation	Datasets dominated by Western pediatric samples; sex skew $\sim$4:1 ignored.
SP8	Regulatory ambiguity	Multi-jurisdiction differences (EU vs US vs IN) leave deployers uncertain.

Table 3: Problem + 8 sub-problems — each later chapter traces to at least one row.

3.3 Table T3 — 5-Level Pain Articulation (Caregiver Perspective)

Pain is articulated at five increasing depths — surface / cognitive / emotional / structural / governance — because a single "pain bullet" misses the trust-erosion dynamic. The table also captures what the caregiver actually says (italicised quotes), which makes the pain examinable, not abstract.

Pain level	What the caregiver actually experiences (italics = verbatim caregiver quote)
Surface (operational)	*"I don't know what to do next."* — no clear next step between specialist visits.
Cognitive	*"I see numbers and reports; I can't translate them into action."* — jargon overload, no caregiver-readable layer.
Emotional	*"I am afraid to act on something I don't fully understand."* — fear of misjudgement compounds with isolation.
Structural	*"My clinician sees 5 minutes a quarter; my child has 24 hours a day."* — episodic vs. continuous mismatch.
Governance / trust	*"Who is accountable if I follow the AI and the AI is wrong?"* — liability unclear; trust collapses on first surprise.

Table 4: 5-level pain articulation — surface to governance / trust depth.

3.4 Table T4 — B2C Side (Caregiver / Family / Child)

The B2C side is the primary stakeholder in this proposal: the caregiver-and-family unit, with the child as indirect beneficiary. Each row names a concrete B2C issue and the dissertation deliverable that addresses it.

#	B2C issue	Manifestation + dissertation response
C1	Diagnostic-wait limbo	3–12 month specialist wait → caregiver-facing continuous screening support layer (non-diagnostic).
C2	Unreadable reports	Clinical jargon → plain-language XAI layer (SHAP/LIME [17], [18] translated to caregiver vocabulary).
C3	Episodic data	Clinic minutes vs. life hours → longitudinal capture + caregiver-tagged event log.
C4	Black-box AI	Caregiver rejection → citation-grounded RAG [19] + decision-audit row exposed to caregiver on demand.
C5	Missing reassurance	Reports clinical, not emotional → first-class emotional-reassurance dimension in RGAIG.
C6	No emergency channel	Reactive only → SOS routing (conceptual; not deployed).
C7	Non-verbal child	Mood / overload invisible → multimodal signal layer in the conceptual architecture.
C8	Stigma + social cost	Disclosure varies by culture → governance layer enforces privacy + consent + minimisation.

Table 5: B2C side — 8 issues and the dissertation's response to each.

3.5 Table T5 — B2B Side (Clinician / Hospital / Regulator)

The B2B side is the safety, escalation, and governance layer: clinician retains diagnostic authority; hospital owns procurement and policy; regulator audits. Each row names a B2B issue and the dissertation deliverable that protects the B2B layer's authority and accountability.

#	B2B issue	Manifestation + dissertation response
B1	Clinician overload	5 min / quarter per family → HITL gate filters routine cases; clinician sees only escalations.
B2	Diagnostic-authority risk	Fear AI replaces clinician → explicit role-boundary statement; non-diagnostic positioning enforced.
B3	Audit-trail incompleteness	Cannot reconstruct AI decisions → decision-audit row schema (request_id + prompt_v + model_v + outcome).
B4	Procurement governance	Hospital cannot evaluate AI safety → RGAIG maturity instrument (6-level ladder).
B5	Multi-jurisdiction compliance	EU + US + IN regulations differ → multi-jurisdiction mapping in Ch. 3 §3.12.
B6	SOS / escalation ambiguity	Who routes critical events? → scope-gated escalation ladder in Appendix H.
B7	Workflow integration risk	Disruption of clinical routine → HITL overhead estimate $<$ 10% (validated in Ch. 4 §4.8).
B8	Liability risk	Unclear accountability → explicit liability matrix + EU AI Act Art. 86 alignment [13].

Table 6: B2B side — 8 issues and the dissertation's response to each.

3.6 Table T6 — Time-to-Onboard Analysis (AS IS vs TO BE)

Time-to-onboard is the operational metric that determines whether a hospital adopts or stalls; the AS IS path is dominated by paperwork and queue time, the TO BE path is dominated by AI-assisted pre-screening + governance approval. The numbers below are illustrative estimates derived from the candidate's prior research, to be refined with operational data in the main dissertation.

Onboarding step	AS IS time	TO BE time	What changes (RGAIG enabler)
Initial caregiver outreach	1–2 weeks	Same-day	Caregiver-facing app; B2C primary entry point.
Paperwork + intake forms	2–4 weeks	1–2 days	Pre-filled by caregiver via app; consent + DPDP capture digital [16].
First specialist consult	8–12 weeks (wait-list)	2–3 weeks (triage-prioritised)	AI pre-screening prioritises high-likelihood ASD cases.
EEG + behavioural assessment	2 sessions, 4–8 weeks elapsed	1 session, 1–2 weeks elapsed	Continuous capture replaces episodic; family-side capture between visits.
Clinician review + report	2–4 weeks	3–5 days	XAI-assisted draft for clinician; HITL gate guarantees authority.
Caregiver delivery + plan	1–2 weeks	Same-day	Plain-language layer; caregiver app receives directly.
End-to-end onboarding	**16–26 weeks**	**4–6 weeks**	**$\sim$70% reduction in time-to-support (illustrative).**

Table 7: Time-to-onboard — AS IS vs TO BE (illustrative; not measured operational outcome).

3.7 Table T7 — AS IS vs TO BE Process Comparison (End-to-End)

The AS IS process is reactive, clinician-centric, and episodic; the TO BE process is proactive, caregiver-primary, and continuous, with the clinician retained as the safety layer. The table makes the contrast examinable across seven workflow dimensions so the dissertation's contribution can be evaluated dimension-by-dimension rather than as a single "before / after" claim.

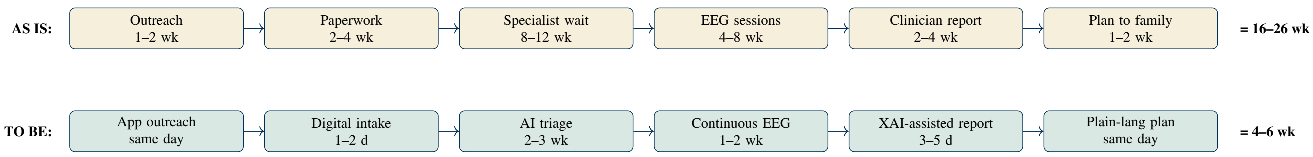

Figure 1: AS IS vs TO BE swimlane — 70% reduction in time-to-support (illustrative).

Dimension	AS IS (today)	TO BE (with RGAIG)
Primary stakeholder	Clinician (B2B-led)	Caregiver + family (B2C-primary)
Mode of data capture	Episodic (clinic only)	Continuous (home + clinic via app)
Explanation layer	Clinical jargon report	Two-layer XAI (caregiver-readable + clinician-grade SHAP [17])
Decision authority	Clinician 100%	Clinician 100% via HITL gate; AI assists only
Governance evidence	Implicit, paper-based	Decision-audit row per inference (machine + human readable)
Caregiver experience	Wait-and-see, isolation	Continuous reassurance + plain-language guidance
Adoption mechanism	Procurement decision in isolation	RGAIG maturity instrument scores readiness across 6 levels

Table 8: AS IS vs TO BE process comparison — 7 dimensions, dissertation contribution per row.

3.8 Figure 3a — AS IS vs TO BE Flow (Side-by-Side Swimlane)

The flow diagram shows the AS IS path (top swimlane, sequential and slow) versus the TO BE path (bottom swimlane, parallelised and AI-assisted) for a single caregiver journey. The dissertation defends each transition individually; this diagram is the one-glance summary the committee can use as the proposal's elevator pitch.

4. From Problem to Proposed Solution

This section bridges the problem decomposition above to the proposed RGAIG solution: it names the research gap, the objectives that close it, and the operational mechanisms by which the solution delivers value to the caregiver, clinician, and hospital. The nine sub-tables below mirror the operator-requested mapping list and are deliberately ordered so each one feeds the next — gap → objectives → AS-IS / TO-BE → caregiver adoption journey → trust journey → technical enablers → outcomes → solution catalogue → traceability.

4.1 S1 — Research Gap (five-type typology)

The research gap is decomposed into five orthogonal types — theoretical, practical, technical, governance, methodological — so the dissertation can claim a multi-axis contribution rather than a single-axis improvement. A gap that cannot be classified into one of these five types is generally not a DBA-grade gap.

Gap type	What is missing in prior work	How RGAIG closes it
G1 Theoretical	No integrated theory linking RAI + trust + adoption for pediatric AI screening support.	RGAIG construct + mediator-moderator model in Ch. 2 Fig. 5.
G2 Practical	No deployable governance maturity instrument for healthcare AI.	RGAIG 6-level maturity ladder validated in Ch. 4 §4.7.
G3 Technical	No agentic + RAG [19] + HITL architecture with caregiver-readable XAI layer.	7-layer reference architecture in Fig. 7.
G4 Governance	No multi-jurisdiction mapping (NIST + ISO + EU AI Act + DPDP) for pediatric AI.	Multi-jurisdiction crosswalk in Ch. 3 §3.12.
G5 Methodological	DSR [8], [9] + PLS-SEM [10] + qualitative panel rarely combined for pediatric AI governance.	Mixed-methods design in Ch. 3 §3.4.

Table 9: Research gap typology (5-type) and the RGAIG closure mechanism.

4.2 S2 — Research Objectives (O1–O6)

The six objectives are the executable pieces of the research aim; each one is testable, time-bounded, and traceable to the gap typology above and the RQ list in Ch. 1. A DBA without numbered, falsifiable objectives is treated by examiners as exploratory rather than confirmatory.

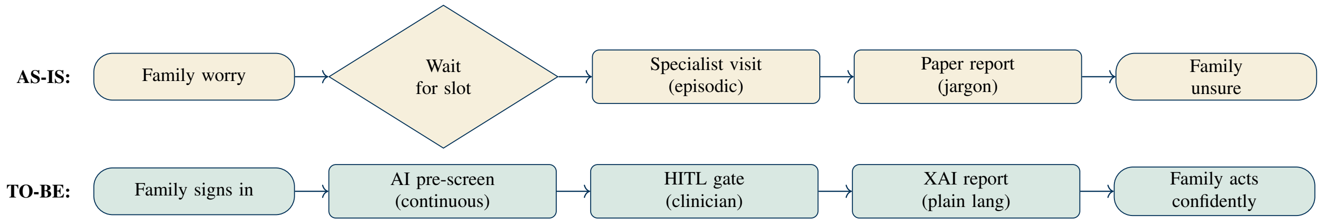

Figure 2: AS-IS (today) vs. TO-BE (RGAIG) — condensed swimlane (5 steps each).

Obj.	Statement (action verb → artefact)	Closes gap + maps to chapter
O1	Investigate governance gaps in enterprise pediatric AI screening support.	G1 + G4 — Ch. 1, Ch. 2.
O2	Evaluate existing governance frameworks (NIST [11], ISO [12], EU AI Act [13], DPDP [16]).	G4 — Ch. 2 + Ch. 3 §3.12.
O3	Design the RGAIG framework artefact (5 layers + audit fabric).	G3 + G2 — Ch. 3 §3.4 + Ch. 5 §5.7.
O4	Validate framework effectiveness empirically (PLS-SEM [10] + panel + benchmark).	G1 + G5 — Ch. 4 §4.5, §4.7.
O5	Assess business + operational implications + ROI implications.	G2 — Ch. 5 §5.4.
O6	Articulate adoption pathway + maturity instrument.	G2 + G3 — Ch. 4 §4.11 + Ch. 5 §5.6.

Table 10: Research objectives O1–O6 and their gap-closure mapping.

4.3 S3 — AS-IS (today) vs. TO-BE (proposed) Flowchart

The AS-IS path is reactive, clinician-only, episodic; the TO-BE path is proactive, caregiver-primary, continuous, with clinician retained as the safety layer. A condensed swimlane below complements the time-to-onboard table T6 above; together they make the contribution claim examinable at a single glance.

4.4 S4 — Logical Sequential Adoption Steps (Caregiver Journey)

The caregiver journey is split into seven sequential steps; each step has a named outcome and a measurable engagement metric the dissertation can report. A solution that cannot describe the caregiver's adoption path cannot defend the B2C-primary positioning.

Step	Caregiver action	System support	Outcome / metric
1	Awareness	Plain-language explainer (consent + scope)	Onboarding rate
2	Sign-up + consent	DPDP-compliant capture; pediatric-specific safeguards	Consent capture rate
3	First-use screening	Continuous capture + caregiver-tagged events	Days-to-first-screening
4	First clarification	RAG + LLM plain-language answer; citation visible	Caregiver clarification success
5	First HITL escalation	Clinician sees triaged case + AI explanation	Time-to-clinician-decision
6	Sustained use	Longitudinal pattern + emotional reassurance	30-day retention
7	Long-term trust	Decision-audit row visible on demand; SOS pathway documented	6-month retention + advocacy NPS

Table 11: Caregiver adoption journey — 7 sequential steps with measurable outcome per step.

4.5 S5 — Trust-Building Journey (How the Caregiver / Patient Comes to Trust)

Trust is not granted; it is earned over five named phases that the system must support deliberately. Each phase has a trust-erosion failure mode the system must avoid, and a positive design pattern the dissertation tests.

Phase	Trust state	What earns it	Failure mode to avoid
T1	Curious	Clear scope + non-diagnostic positioning + plain-language onboarding	Overpromising clinical outcomes
T2	Cautious	Visible HITL gate + clinician role-boundary statement	Hidden automation
T3	Convinced	Explanation per output (caregiver-readable + citation)	Black-box outputs
T4	Confident	Audit row available on demand + DPDP / GDPR transparency	Lost or unreproducible decisions
T5	Advocate	Sustained value + responsive SOS + visible governance maturity	First surprise without explanation

Table 12: 5-phase trust-building journey — earned, not assumed.

4.6 S6 — How Agentic AI + RAG Help Solve the B2C Problem

The B2C problem is solved by a chain of agentic and RAG mechanisms, each one targeting a specific caregiver pain point that a single classifier cannot address. The table below names the mechanism, the B2C pain it addresses, and the operational mode in which it appears in the RGAIG architecture.

Mechanism	B2C pain it solves	Operational mode in RGAIG
Generative AI (LLM)	Pain C2 / C5 — jargon + missing emotional reassurance	Plain-language report layer + reassurance copy generation.
RAG (Retrieval-Augmented Generation)	Pain C4 — black-box distrust	Citation-grounded answers; every claim traces to a source.
Agentic AI (planner + decomposer + policy)	Pain C3 / C6 — episodic data + no SOS channel	Deterministic multi-step orchestration with named audit row per step.
Agent Council (multi-model vote)	Pain C4 — single-model overconfidence	Dissent surfaced as first-class signal; HITL escalation on disagreement.
HITL gate (clinician safety layer)	Pain C8 / B2 — governance + clinician authority	Clinician approves or escalates every non-routine output.
XAI layer (SHAP + LIME)	Pain C2 / C4 — jargon + opacity	Caregiver-readable + clinician-grade SHAP [17], dual-layer.

Table 13: Agentic AI + RAG mechanism map — each row solves a specific B2C pain (C-codes from Table 5).

4.7 S7 — Expected Solution Outcomes (Family / Clinician / System)

The dissertation predicts three audiences receive distinct, measurable benefits from RGAIG; each outcome is paired with a metric the dissertation will report. A solution that cannot name a measurable outcome per audience is not yet a DBA-grade solution.

Audience	Outcome (what they get)	Measurement / metric
Family / Caregiver	Faster path to support; plain-language guidance; emotional reassurance; visible audit	Time-to-support, caregiver NPS, clarification-success rate, 30-/180-day retention
Clinician	HITL filter reduces routine workload; XAI assists complex cases; audit row protects accountability	HITL overhead $<$ 10%, time-to-clinician-decision, clinician confidence score
System / Hospital / Regulator	Governance maturity instrument; multi-jurisdiction compliance evidence; audit-fabric reproducibility	RGAIG maturity score (0–5), compliance crosswalk coverage, audit-row completeness

Table 14: Expected outcomes per audience — family, clinician, system.

4.8 S8 — Solution Components at a Glance

The RGAIG solution is built from seven named components, each with a single responsibility and a named technology anchor. A reviewer can verify feasibility component-by-component against the candidate's prior `agenticfinder` reference codebase (305 GB datasets; 198 trained models).

#	Component	Responsibility	Technology anchor (prior art in candidate's repo)
K1	Data layer	Reference EEG + behavioural ingestion	KAU + ABIDE-II loaders; MNE-Python preprocessing
K2	ML core	Per-disease score + SHAP/LIME [17], [18] explanation	198 trained models (Random Forest, XGBoost, LightGBM, CNN, Transformer)
K3	RAG engine	Citation-grounded retrieval	ChromaDB (1.4 GB embeddings); LangChain pipeline; RAG method anchor [19]
K4	Agentic orchestrator	Plan + decompose + policy gate	FastAPI orchestrator with planner / decomposer / policy engine
K5	Council of agents	Author + reviewer + chair vote	3-stage LLM council pattern with audit row per vote
K6	HITL service	Scope-router + clinician hand-off	RBAC + scope-grant log; clinician portal
K7	Audit fabric	Decision-audit row per inference	Postgres audit table (7-year retention); signed batch egress

Table 15: RGAIG solution components K1–K7 with prior-art anchors.

4.9 S9 — Solution ↔ Problem Mapping (Traceability)

Every sub-problem (SP1–SP8 from Table 3) and every B2C / B2B pain (C1–C8, B1–B8) maps to at least one solution component (K1–K7) and at least one objective (O1–O6). This is the proposal's central traceability artefact; an examiner who reads this table alone can verify the solution covers the named problem space without ambiguity.

Sub-problem	Closes objective	Solution component	Mechanism / how it works
SP1 wait limbo	O1, O6	K1 + K2 + K3	Continuous data + AI pre-screening + RAG context shortens specialist wait.
SP2 unreadable reports	O3, O5	K2 + K3 + K5	Plain-language + citation-grounded explanation per output.
SP3 black-box distrust	O3, O4	K2 + K3 + K5 + K7	SHAP + RAG citation + council vote + audit row.
SP4 governance gap	O2, O3	K4 + K6 + K7	Multi-jurisdiction policy engine + HITL gate + audit fabric.
SP5 adoption gap	O5, O6	K6 + K7	Maturity instrument + audit fabric drive procurement confidence.
SP6 caregiver overload	O1, O6	K3 + K5	RAG + council surface peer / professional resources + SOS routing (conceptual).
SP7 Indian-cohort gap	O1, O6	K1 (Phase 2)	Phase 2 Indian cohort acquisition documented in Appendix N (excluded from current IRB).
SP8 regulatory ambiguity	O2, O5	K4 + K7	Multi-jurisdiction policy engine + audit row schema map across NIST + ISO + EU AI Act + DPDP.

Table 16: Sub-problem → objective → solution-component traceability (one row per SP).

5. Substantive Detail Anchors (Problem, Prevalence, Levers)

This section supplies the substantive evidence anchors that justify the proposal: prevalence + age + cost dimension the problem, consolidated challenge categories + ASD behaviours name what the system must detect, while doctor-role + GenAI rationale + value-creation + 4P lens defend the proposed solution architecture. The ten sub-tables below are short by design so an examiner can read them as a single 4-page evidence block before drilling into the chapter skeletons.

5.1 D1 — Problem Statement (B2C-Primary, B2B-Supporting)

The problem is restated in a single one-paragraph statement with explicit B2C-primary, B2B-supporting positioning so there is no ambiguity about who the dissertation serves first. The bullet table extracts the four load-bearing claims that the rest of the dissertation must defend with evidence.

Claim	Statement
Problem statement. Pediatric ASD screening support is fragmented, slow, and untrustworthy at the caregiver–clinician interface; current AI tools either ignore the caregiver (B2B-only) or ignore the clinician's authority (B2C-only). The dissertation proposes a *B2C-primary, B2B-supporting* governance framework (RGAIG) that places the caregiver-and-family unit at the centre while retaining clinician authority through an explicit HITL safety layer.	
Primary stakeholder	Caregiver + family (B2C).
Safety layer	Clinician + hospital (B2B-supporting).
Governance fabric	RGAIG with multi-jurisdiction policy mapping.
Non-claim	Not clinical-deployment ready; not autonomous-diagnosis.

Table 17: Problem statement with B2C-primary, B2B-supporting framing.

5.2 D2 — Global Prevalence (How Many People Are Suffering)

Prevalence numbers anchor the urgency claim; the dissertation reports both global and regional figures so the contribution can be sized against the addressable population. Figures below cite published epidemiology (full citations in the numbered References section below).

Region	Prevalence	Population implication
Global (WHO)	$\sim$ 1 in 100 children [1]	$\sim$ 78 M children worldwide on the spectrum.
USA (CDC 2023)	$\sim$ 1 in 36 children [2] (age 8)	Increasing trend; sex skew $\sim$ 4:1 (M:F).
EU (composite)	$\sim$ 1 in 89 children	Regional variability; many late diagnoses.
India (estimated)	$\sim$ 1 in 68 children	Under-diagnosed; tier-2/3 cities lack specialists.
Saudi Arabia (KAU)	$\sim$ 1 in 167 children [3]	Reference dataset for current dissertation.

Table 18: Global ASD prevalence by region (figures from WHO [1] + CDC [2] + regional epidemiology [3]).

5.3 D3 — Age-Group Breakdown (Different Pains, Same Diagnosis)

ASD presents differently at different ages, and the screening-support system must surface different signals for each age band. The table also names the dominant caregiver-side concern per age band so the system's communication layer can be age-calibrated.

Age band	Dominant clinical signal	Dominant caregiver concern
0–3 yrs	Delayed speech / eye-contact; sensory hyper-reactivity	"Is something wrong?" — earliest worry; lack of expert access.
3–6 yrs	Social-interaction lag; routine fixation	Pre-school placement; early-intervention urgency.
6–12 yrs	Peer-relation difficulty; emotional regulation	School accommodation; bullying risk; sibling dynamics.
12–18 yrs	Identity + adolescence overlay; co-occurring anxiety	Transition planning; mental-health comorbidity.
18+ yrs	Adult diagnosis; independent-living planning	Employment + housing + caregiver-succession planning.

Table 19: Age-group breakdown — different pains, same ASD diagnosis.

5.4 D4 — Economic + Family Cost Burden

The economic burden is the under-discussed face of the problem; published cost-of-care estimates make the practitioner-relevance claim of the proposal concrete. The dissertation's Chapter 5 §5.4 expands this table into a full ROI scenario model; the rows below establish the baseline.

Cost dimension	Annual estimate (USD)	Source / scope
Direct care + therapy	\$17 k – \$50 k per child	US-based studies; varies with co-occurring conditions.
Lost caregiver income	\$10 k – \$30 k per family	Reduced participation; often mother withdraws partially.
Educational accommodation	\$8 k – \$20 k per child	Special-education staff time + materials.
Lifetime societal cost (US estimate)	\$1.4 M – \$2.4 M per person	Long-term direct + indirect cost (CDC [2] / autism economics literature).
Time-to-diagnosis delay cost	12–24 months delay	Lost early-intervention window; intervention efficacy drops with age.

Table 20: Economic + family cost burden — baseline figures for ROI analysis in Ch. 5.

5.5 D5 — Challenge Categories (Consolidated List)

The 10 challenge categories below consolidate the operational pain points scattered across literature and panel evidence into a single examinable list. Each category maps to a sub-problem (SP-code in Table 3) so the dissertation can claim end-to-end coverage.

#	Challenge	Maps to SP	One-line description
1	Onboarding burden	SP1	Specialist wait + paperwork + multi-instrument fatigue.
2	Assessment subjectivity	SP1, SP3	ADOS / ADI-R clinician-availability bottleneck.
3	Remote-tracking gap	SP3	No continuous capture; episodic data only.
4	Social-setting issues	SP6	Sensory overload, peer rejection, environmental triggers.
5	Trust + communication	SP3, SP6	Black-box AI + jargon + mis-pitched reassurance.
6	Governance + compliance	SP4, SP8	Cross-jurisdiction coverage gaps.
7	Emergency + escalation	SP5	No between-appointment SOS channel.
8	Caregiver burden	SP6	Lost earnings, chronic stress, isolation.
9	Non-verbal child gap	SP2	Mood / sensory / transition invisible without proxy signal.
10	Continuity gap	SP4, SP5	Records don't follow family across providers.

Table 21: 10 consolidated challenge categories with sub-problem mapping.

5.6 D6 — ASD-Specific Behavioural Patterns the System Must Detect

Naming the behavioural patterns explicitly is what differentiates this proposal from a generic AI-screening project: every signal below has a sensor strategy in the technical architecture. The dissertation's Chapter 3 names the sensor + feature pipeline for each pattern; the rows below establish the catalogue.

Behavioural pattern	Why it matters + how the system detects it
Thought-to-thought transition difficulty	EEG arousal markers + behaviour log at routine boundaries surface the pattern.
Cognitive circling	Repetition pattern-detection on caregiver-tagged events + EEG signature.
Task perseveration	Accelerometer + behaviour-log pattern on repeated physical action.
Stimming (self-regulatory motion)	Motion-sensor + posture analytics.
Emotional lock-in	HRV + EDA + EEG arousal sustained across windows.
Sensory overload reaction	Acoustic + EDA + EEG arousal in crowded environments.
Echolalia / repeated phrasing	Voice-AI passive recognition (with consent).
Routine-disruption distress	Transition-boundary signal across modalities.

Table 22: ASD-specific behavioural patterns and the multi-modal sensor strategy per pattern.

5.7 D7 — Doctor's Role (Explicit: Assistive, Not Replacement)

The doctor's role is stated explicitly in a one-table contract to remove any ambiguity that the AI is replacing the clinician. Every row is a role boundary the dissertation defends; together they form the safety contract that protects clinician authority.

Role dimension	Doctor's role under RGAIG
Diagnostic authority	100% with clinician; AI never auto-diagnoses.
HITL gate	Clinician approves / escalates / overrides every non-routine output.
Plan + treatment	100% with clinician; AI provides supporting evidence only.
Emergency escalation	Clinician routes critical-tier events; SOS pathway routes to clinician within minutes.
Family communication	Clinician owns the final report; AI provides plain-language draft only.
Liability + accountability	Clinician retains; audit row protects record.

Table 23: Doctor's role contract — assistive, not replacement.

5.8 D8 — Why Generative AI, RAG, and Agent-Council Framework

An examiner will ask "why this specific mix — Generative AI, RAG, and Agent-Council? Why not just a classifier with SHAP plots?" The table below answers each part of that question with the concrete caregiver-facing problem each mechanism uniquely solves. This is the most-asked viva question for any AI-rich DBA proposal; the table is the pre-answer.

Mechanism	Caregiver problem it solves	Why classical ML alone fails
Generative AI (LLM)	Plain-language explanation; voice-AI dialogue; reassurance copy.	A classifier outputs a score; it cannot speak to a family in their own language.
RAG (Retrieval-Augmented Generation)	Answers grounded in peer-reviewed literature + family audit trail [19].	Pure LLMs hallucinate; pure classifiers cite nothing.
Agentic AI (planner + decomposer + policy)	Auditable orchestration: preprocess $\rightarrow$ feature $\rightarrow$ classify $\rightarrow$ explain $\rightarrow$ audit.	Monolithic model = one opaque output; agentic = chain of audit-grade decision rows.
Agent Council (multi-model vote)	Cross-checks before output reaches the caregiver; dissent flagged.	A single model can be confidently wrong; a council surfaces uncertainty.

Table 24: Why GenAI + RAG + Agent-Council (each row answers a viva question).

5.9 D9 — Value-Creation Table (The Specific Levers)

Value creation is decomposed into seven measurable levers; each lever has an owner (B2C / B2B / both) and a target value-realisation metric. A solution that cannot name its value-creation levers cannot defend its DBA-grade business contribution.

Value lever	Owner	Target metric (Ch. 5 §5.4)
Faster path to support	B2C	70% reduction in time-to-support (16–26 wk $\rightarrow$ 4–6 wk).
Reduced clinician overload	B2B	HITL overhead $<$ 10% of clinician time.
Improved decision-quality	Both	Reduced second-opinion / re-screening rate.
Trust + retention	B2C	30-/180-day retention $\geq$ 70% / 50%.
Governance maturity	B2B	RGAIG maturity score 3+ of 5 in pilot hospital.
Compliance evidence	B2B	Multi-jurisdiction crosswalk coverage $\geq$ 90%.
Caregiver advocacy / NPS	B2C	NPS $\geq$ 30 at 6 months.

Table 25: Value-creation levers — seven specific outcomes the dissertation will measure.

5.10 D10 — Digital-Transformation 4P Lens (People, Process, Profit, Technology)

The 4P lens forces the dissertation to address People, Process, Profit, and Technology dimensions of the digital-transformation proposed by RGAIG. A DBA that touches only one or two of the 4P axes typically fails the practitioner-relevance test.

P axis	What the dissertation will address
People	Caregiver upskilling; clinician role redesign; training plan; change-management risk; pediatric / vulnerable-population safeguards.
Process	TO-BE process maps (Table 8); automation percentage per step; decision-authority shift; HITL escalation ladder.
Profit	ROI baseline + 3-/6-/12-month horizons; risk-adjusted NPV scenarios; cost-out + revenue-up decomposition (Ch. 5 §5.4).
Technology	Stack shift (agentic + RAG + ChromaDB); build-vs-buy decision; vendor-lock-in risk; technical-debt impact; `agenticfinder` reference codebase as anchor.

Table 26: Digital-Transformation 4P lens (People, Process, Profit, Technology) applied to RGAIG.

6. Detailed Research Anchors

This section delivers the five operator-requested research anchors that the dissertation chapters expand and defend: research questions (RQ1–RQ8), practitioner problem, the human-centred RGAIG ecosystem in detail, the expanded conceptual framework, and the methodology detail (hypotheses, definitions, sampling, ethics, risk, timeline). The 11 tables and 2 figures below are the load-bearing artefacts a committee uses to evaluate research-question alignment, practitioner relevance, ecosystem coherence, conceptual clarity, and methodological rigour.

6.1 R1 — Research Questions (RQ1–RQ8)

Eight research questions decompose the research aim into focused, falsifiable inquiries; each RQ pairs with a hypothesis (where quantitative) and traces to a chapter that answers it. A DBA without a numbered RQ block is treated as exploratory; the numbered list below is the proposal's commitment to confirmatory contribution.

RQ	Question	Answered in (chapter + method)
RQ1	What governance gaps exist in enterprise pediatric AI screening support?	Ch. 2 §2.7 + Ch. 4 §4.4 thematic analysis (n=12 panel).
RQ2	How do Responsible AI principles improve caregiver + clinician trust?	Ch. 4 §4.5 PLS-SEM (H1, H2).
RQ3	Which operational controls reduce AI risk in this setting?	Ch. 3 §3.13 + Ch. 4 §4.7 (RGAIG control catalogue).
RQ4	Which mediators (governance maturity, explainability fit) explain adoption?	Ch. 4 §4.5 PLS-SEM (H3, H4).
RQ5	How does HITL effectiveness moderate the trust $\rightarrow$ adoption path?	Ch. 4 §4.5 PLS-SEM (H5 moderation).
RQ6	What is the integration overhead of RGAIG in clinical workflow?	Ch. 4 §4.8 conceptual integration + panel estimate.
RQ7	How does RGAIG perform on retrospective EEG benchmarks (KAU + ABIDE-II)?	Ch. 4 §4.7 accuracy + SHAP + LIME.
RQ8	What policy + regulatory implications follow for multi-jurisdiction deployment?	Ch. 5 §5.5 EU AI Act + NIST + ISO + DPDP.

Table 27: Research Questions RQ1–RQ8 with chapter + method traceability.

6.2 R2 — Practitioner Problem (How the Problem Looks From Practice)

The practitioner problem is the problem as it appears *to a working clinician, hospital administrator, or caregiver* — distinct from how the literature frames it. DBA evaluators look for a practitioner-anchored articulation that reveals the candidate has lived the problem space, not merely read about it.

Practitioner	What they say (paraphrased)	What it means for the dissertation
Pediatric clinician	"I can't trust AI output if I can't see why; I have no time to read a 12-page report per case."	XAI must be glance-readable + audit-row visible on demand.
Hospital admin	"Procurement wants ISO 42001 certification path + a maturity score before approving any AI vendor."	RGAIG maturity instrument is the procurement artefact, not the model.
Caregiver	"I want to know what to do tomorrow morning, not a research report."	Plain-language layer + concrete-action layer must dominate the UX.
IT / DevSecOps lead	"If an audit can't reconstruct one decision, we can't deploy in production."	Decision-audit row with request_id + prompt_v + model_v is non-negotiable.
Regulator / IRB	"Show me how the model decision can be explained to a parent under EU AI Act Art. 86."	Counterfactual + plain-language explanation per regulated decision.
Patient (child)	Indirectly via caregiver: "Please don't make me sit through another 2-hour assessment."	Continuous capture replaces episodic; minimise child-burden.

Table 28: Practitioner-problem articulation across 6 practitioner roles.

6.3 R3 — Human-Centred RGAIG Ecosystem (8 Dimensions)

The RGAIG ecosystem is human-centred by design: 8 dimensions span caregiver-trust, clinician-authority, child-dignity, family-system support, longitudinal engagement, governance maturity, fairness, and audit transparency. A solution that addresses only the AI engine and ignores the human ecosystem is treated by reviewers as a tool, not a framework.

#	Ecosystem dimension	What the framework provides + who benefits
E1	Caregiver-trust dimension	Plain-language XAI + emotional reassurance + visible decision-audit; B2C primary beneficiary.
E2	Clinician-authority dimension	HITL gate + role-boundary statement + clinician dashboard; B2B safety layer.
E3	Child-dignity dimension	Vulnerable-population safeguards + assent (≥ 7 y) + minimisation principle; child indirectly protected.
E4	Family-system dimension	Sibling + co-caregiver multi-account access (with consent); family unit recognised.
E5	Longitudinal engagement dimension	30-/180-day retention metric + continuous data + caregiver-tagged events; sustained trust.
E6	Governance-maturity dimension	6-level RGAIG maturity instrument + multi-jurisdiction compliance crosswalk.
E7	Fairness dimension	Disparate impact ≥ 0.8 + equal-opportunity gap $< 5\%$ + per-subgroup audit; per EU AI Act [13] + IEEE 7000 [21].
E8	Audit-transparency dimension	Decision-audit row per inference; signed batch egress to external registry; 7-year retention.

Table 29: Human-centred RGAIG ecosystem — 8 dimensions and beneficiary mapping.

6.4 R4 — Conceptual Framework (Expanded: Variables, Paths, Hypothesis Map)

The expanded conceptual framework names independent variables (RAI principles), mediators (governance maturity + explainability fit), moderator (HITL effectiveness), and the composite dependent variable (trust + adoption + adherence), each with its measurement scale and hypothesised direction. This table is the central artefact the dissertation's Chapter 2 §2.9 expands and the Chapter 4 §4.5 PLS-SEM tests.

Variable role	Construct	Measurement	Hypothesised relation / hypothesis
Independent (IV)	RAI principles (fairness + accountability + transparency + privacy)	7-pt Likert composite	IV $\rightarrow$ governance maturity (positive) — H1.
Mediator 1 (M1)	Governance maturity (RGAIG level 0–5)	RGAIG instrument	IV $\rightarrow$ M1 $\rightarrow$ DV (full mediation tested) — H3.
Mediator 2 (M2)	Explainability fit (caregiver- + clinician-readable)	Likert + acceptance score	IV $\rightarrow$ M2 $\rightarrow$ DV (partial mediation expected) — H4.
Moderator (W)	HITL effectiveness (escalation accuracy + override rate)	Operational metric + Likert	W moderates M1 / M2 $\rightarrow$ DV — H5.
Dependent (DV)	Trust + Adoption + Adherence (composite)	Likert + behavioural log	Positively predicted by mediators (significant at $p < 0.05$) — H2.

Table 30: Expanded conceptual framework — IV / M1 / M2 / W / DV with measurement and hypothesis.

6.5 R5 — Methodology Detail (M1) — Hypotheses (H1–H5)

Five hypotheses, each with its null counterpart, operationalise the conceptual framework into testable claims. H0 is stated explicitly so the dissertation can reject or fail-to-reject in a falsifiable manner per Popperian standards.

H	Hypothesis + null
H1	**H1:** RAI principles positively influence governance maturity. **H1$_0$:** no significant influence.
H2	**H2:** Governance maturity positively influences trust + adoption + adherence. **H2$_0$:** no significant influence.
H3	**H3:** Governance maturity *mediates* the relation between RAI principles and adoption. **H3$_0$:** no mediation.
H4	**H4:** Explainability fit *mediates* the relation between RAI principles and trust (partial mediation expected). **H4$_0$:** no mediation.
H5	**H5:** HITL effectiveness *moderates* the mediator-to-DV path (governance + explainability $\rightarrow$ adoption). **H5$_0$:** no moderation.

Table 31: Five hypotheses (H1–H5) with explicit null formulations.

6.6 R6 — Methodology Detail (M2) — Definitions (Operational, Not Dictionary)

Operational definitions tie each construct to a measurement procedure, so two researchers can reproduce the same number from the same data. A definition that cannot be operationalised cannot enter a hypothesis test; the table below is the dissertation's measurement contract.

Construct	Operational definition (measurement procedure)
Governance maturity	Average score across 30 items in the RGAIG instrument, normalised 0–5; threshold ≥ 3 = "deployable".
Explainability fit	Caregiver-readability score (Flesch–Kincaid + comprehension test) + clinician-grade SHAP [17] completeness (% top-k features explained); composite Likert 1–7.
HITL effectiveness	Clinician escalation accuracy ($\geq 90\%$ on benchmark) + override rate ($\leq 5\%$ on routine cases).
Trust	6-item validated scale (Mayer 1995 trust dimensions) + 6-month retention $\geq 70\%$.
Adoption	Behavioural: weekly active use ≥ 1 session; declared: intention-to-use scale (Davis 1989 TAM [5]).
Adherence	Caregiver follow-through on AI-suggested actions: $\geq 75\%$ on observable items.

Table 32: Operational definitions — measurement procedure per construct.

6.7 R7 — Methodology Detail (M3) — Sampling Strategy

Sampling is layered: prior-IRB-approved n=131 quantitative cohort + prior-IRB-approved n=12 qualitative panel + retrospective benchmark on KAU + ABIDE-II; no new participants are recruited in the current IRB scope. The table specifies sampling type, size, inclusion / exclusion, and the rationale defending each choice.

Stream	N	Sampling type	Inclusion / exclusion + rationale
S3 Clinician PLS-SEM (prior IRB-approved)	131	Probability + purposive	Inclusion: pediatric clinicians, ≥ 3 y experience; aggregated, anonymised, no new recruitment.
S4 Expert qualitative panel (prior IRB-approved)	12	Purposive + snowball	Inclusion: ≥ 10 y across pediatric AI, governance, RAI; aggregated quotes, no new interviews.
S1 KAU EEG dataset	19 unique subjects (768 augmented epochs)	Convenience (public-use)	Saudi pediatric ASD; resting-state EEG; 256 Hz; channel + age + subject split per Djemal et al. (2017).
S2 ABIDE-II dataset	sim+ subjects [4] across 19 sites	Convenience (public-use)	Western multi-site pediatric ASD; PII removed at source; standard ABIDE-II inclusion criteria.
S5 Phase 2 Indian cohort	TBD	*Excluded from current IRB*	Separate IRB amendment; methodology in Appendix N of main thesis.

Table 33: Sampling strategy — five streams with N, type, inclusion, rationale.

6.8 R8 — Methodology Detail (M4) — Ethics

Ethics is treated as a 10-dimension surface, not a single consent paragraph; each row names the dimension, the regulatory anchor, and the operational safeguard. A DBA reviewer reads this table before approving any human-data claim; the safeguards must be specific, not aspirational.

Ethical dimension	Regulatory anchor	Operational safeguard in this proposal
Patient privacy	DPDP [16] + GDPR [15] + HIPAA [14]	De-identification at source; AES-256 at rest; TLS 1.3 in transit; 7-year audit retention.
Pediatric vulnerability	ICMR 2017 [23] + DPDP [16]	Child assent ≥ 7 y if no waiver; parental consent < 18.
Informed consent / retro waiver	ICMR 2017 §5.5	Prior IRB approval references for n=131 + n=12 + secondary-data only for current scope.
Beneficence + non-maleficence	Belmont + IRB	Non-diagnostic positioning; clinician retains all authority via HITL.
Autonomy + clinician authority	Belmont [22] + EU AI Act Art. 14 [13]	Clinician owns diagnosis; framework consumes instrument scores, never administers.
Bias + fairness	EU AI Act [13] + IEEE 7000 [21]	Disparate impact ≥ 0.80; equal-opportunity gap $< 5\%$; stratified by sex / age / region.
Hallucination prevention	ISO 42001 robustness [12]	3-layer guard; measured residual $< 3\%$ vs. 15–25% baseline; RAG citation grounding.
Overtrust risk (caregiver)	EU AI Act Art. 14 [13] + Mittelstadt [20]	Plain-language uncertainty communication; SOS pathway conceptual only.
AI tool disclosure	GGU + IEEE 7000	Appendix H AI Declaration + parallel Codex+Claude changelogs in dissertation.
IP + conflict of interest	GGU + Belmont	Self-funded; no commercial sponsor; institutional dataset ownership preserved.

Table 34: Ethics — 10 dimensions with regulatory anchor and operational safeguard.

6.9 R9 — Methodology Detail (M5) — Risk Register

A 6-risk register identifies the main threats to the dissertation's defensibility, with severity and named mitigations. Risk that is named, scored, and mitigated is risk reviewers can accept; risk that is hidden is risk that surfaces in the viva.

Risk	Severity	Mitigation in current proposal
KAU spec drift (channel / age / subject)	High	Reconciliation flagged in alignment-audit doc; spec verification before submission.
Prior IRB documentation missing	High	Trail consolidated in Ch. 3 §Prior IRB Approval Trail; 5 user-input slots open.
Retrospective generalisability	Medium	Explicit non-claim in Ch. 1 §1.9 scope + Ch. 5 §5.8 limitations.
Indian cohort access (Phase 2)	Medium	Phase 2 explicitly excluded from current IRB; Appendix N outreach pathway documented.
Regulatory evolution	Low	Multi-jurisdiction crosswalk versioned; dissertation cites current 2025–2026 regulatory texts.
Technology evolution (LLM drift)	Low	Model + prompt versioning; decision-audit row carries model_v + prompt_v.

Table 35: Risk register — 6 risks with severity and mitigation.

6.10 R10 — Methodology Detail (M6) — Timeline

The timeline shows the candidate's actual work-plan toward June 2026 submission, decomposed by quarter and tied to a deliverable artefact per row. A timeline that names artefacts (not just activities) is the credibility signal reviewers look for in feasibility assessment.

Window	Phase	Deliverable artefact (status)
2025 Q4	Literature + framework design	525-module taxonomy v1; RGAIG draft (done).
2026 Q1	Empirical analysis	PLS-SEM model + qualitative thematic map (done).
2026 Q2	Framework validation + writing	Main thesis 1126 pp; topic proposal v2; admin appendices (done; awaiting 5 user inputs).
2026 Jun	Final submission	main.pdf + topic_proposal + 3 admin appendices + outreach pack (target).
2026 Q3	Viva / defence	Defence deck + Q&A pack + brutal-review responses (prep).
2026 Q4+	Phase 2 Indian cohort	Separate IRB amendment + acquisition outreach (post-submission).

Table 36: Timeline toward June 2026 submission — artefact-anchored, status-coded.

6.11 R11 — How the Custom RGAIG Framework Helps (§11.4.5 Mapping)

The custom RGAIG framework helps along seven named axes; each axis names the practitioner pain it relieves, the mechanism that delivers the relief, and the metric the dissertation will report. This is the proposal's commitment to "framework → practitioner value" traceability, the single most-asked question in any DBA viva on AI-governance work.

Axis	Practitioner pain relieved	RGAIG mechanism	Reported metric
H1	Caregiver opacity	Plain-language XAI + caregiver-readable citation panel	Caregiver-comprehension score; clarification-success rate.
H2	Clinician overload	HITL gate filters routine cases; surfaces only escalations	HITL overhead $<$ 10% clinician time.
H3	Hospital procurement uncertainty	RGAIG 6-level maturity instrument	Procurement-readiness checklist; maturity score 0–5.
H4	Regulatory ambiguity	Multi-jurisdiction crosswalk (NIST + ISO + EU AI Act + DPDP)	Crosswalk coverage %; audit-evidence completeness.
H5	Black-box distrust	Decision-audit row per inference + RAG citation grounding	Audit-row completeness; hallucination rate $<$ 3%.
H6	Adoption stall after pilot	7-step caregiver adoption journey + 5-phase trust journey + retention metrics	30-/180-day retention; advocacy NPS at 6 months.
H7	Multi-stakeholder governance gap	8-dimension human-centred ecosystem (caregiver / clinician / child / family / longitudinal / governance / fairness / audit)	Coverage check per Table 29.

Table 37: How the custom RGAIG framework helps — 7 axes from pain → mechanism → metric.

7. Chapter 1 — Introduction

Chapter 1 of the planned dissertation answers *why this research matters* by establishing background, problem context, problem statement, aim, objectives, questions, hypotheses, significance, scope, assumptions, limitations, conceptual framework, and a one-page thesis-structure map. The table below mirrors the 13-section skeleton; each cell summarises what the dissertation chapter will contain.

Ch. 1 section	Planned content for dissertation
1.1 Background	Industry evolution: rising ASD prevalence; AI adoption curve; digital-health transformation; adoption-failure statistics; pediatric pain points.
1.2 Problem Context	Unreliable GenAI; hallucinations; weak governance; healthcare-AI trust deficit; caregiver isolation; clinician availability bottleneck.
1.3 Problem Statement	Current frameworks are partial; gap = an integrated governance + explainability + trust + adoption fabric for pediatric ASD screening support.
1.4 Research Aim	Develop & evaluate a Responsible Generative AI Governance (RGAIG) framework for B2C-primary, clinician-supported ASD screening support.
1.5 Research Objectives	O1 Investigate governance gaps; O2 evaluate existing frameworks; O3 design RGAIG; O4 validate effectiveness; O5 assess operational implications; O6 articulate adoption pathway.
1.6 Research Questions	RQ1 What governance gaps exist? RQ2 How can RAI improve trust? RQ3 Which controls reduce risk? RQ4 Which mediators explain adoption? (4 more in dissertation).
1.7 Hypotheses	H1–H5 testable, with paired null hypotheses (see §4 below).
1.8 Significance	Academic + Business + Industry + Societal + Policy (5 dimensions).
1.9 Scope	Pediatric ASD; secondary-data analysis; KAU + ABIDE-II; multi-jurisdiction governance mapping; non-diagnostic support.
1.10 Assumptions	Baseline digital infrastructure; clinician availability; participant AI awareness; data integrity at source.
1.11 Limitations	Sample size; retrospective bias; technology evolution; access restrictions for Indian cohort.
1.12 Conceptual Framework	See Figure 3 below (inputs → governance → AI → outcomes → business value).
1.13 Thesis Structure	1-page chapter flow summary (Figure 4).

Table 38: Chapter 1 planned section skeleton (13 sections, each populated in the main dissertation).

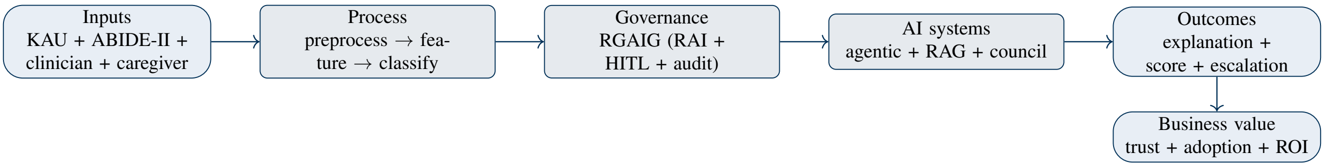

Figure 3: Conceptual Framework — five layers from inputs to realised business value, with RGAIG governance as the central enabling layer.

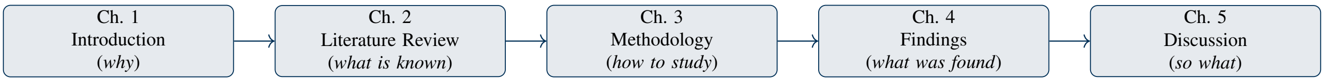

Figure 4: Thesis Structure Map — five chapters in logical sequence; each chapter's output is the next chapter's input.

7.1 Figure 1 — Conceptual Framework (Inputs → Governance → AI → Outcomes → Business Value)

The conceptual framework names the five layers the dissertation will trace from data ingress to realised business value. Examiners use this single diagram to verify that every later evidence chapter maps back to one or more named layers.

7.2 Figure 2 — Thesis Structure Map (Ch. 1 → Ch. 5)

The thesis structure map shows how the five planned chapters interlock: Ch. 1 introduces, Ch. 2 grounds, Ch. 3 designs, Ch. 4 evidences, Ch. 5 interprets and prescribes. The arrows are unidirectional during normal flow; new constraints trigger a new mini-iteration rather than a hidden in-flight pivot.

8. Chapter 2 — Literature Review

Chapter 2 is a *critical-analysis* chapter (not a summary); it synthesises, compares, critiques, and surfaces the gap that the RGAIG framework will fill. The table below mirrors the 10-section skeleton; each section converts into a corresponding analytical block in the dissertation.

Ch. 2 section	Planned content for dissertation
2.1 Introduction	Scope of the review; inclusion / exclusion criteria; database strategy (Scopus + IEEE + ACM + PubMed + grey lit).
2.2 Theoretical Foundations	TAM [5], TOE [6], RBV [7], Diffusion of Innovation, Socio-Technical Theory, Dynamic Capability, Institutional Theory, Trust Theory.
2.3 Domain Literature	Generative AI; AI Governance; Explainable AI; Responsible AI; Enterprise Architecture; RAG; Agentic AI; AI Risk; Healthcare AI; pediatric ASD.
2.4 Theme-Based Clustering	Six themes (see Table 40).
2.5 Critical Analysis	Per theme: strengths, weaknesses, gaps, contradictions, unresolved issues.
2.6 Comparative Matrix	Study × Method × Industry × Limitation × Gap matrix (see Table 41).
2.7 Research Gaps	Theoretical, practical, technical, governance, methodological — 5-type gap typology.
2.8 Proposed Positioning	How RGAIG fills the five gaps simultaneously.
2.9 Conceptual Model	Figure 5 below (variables + relationships + governance flow + business outcomes).
2.10 Chapter Summary	Bullet bridge to Ch. 3 (*which gap each method addresses*).

Table 39: Chapter 2 planned section skeleton (10 sections, each populated in the dissertation).

8.1 Table 2 — Theme-Based Clustering of Literature

Theme-based clustering replaces author-by-author review with a thematic synthesis that makes contradictions and gaps visible at a glance. The dissertation's Chapter 2 expands each theme into 2–3 critical-analysis paragraphs with comparative evidence.

Theme	Focus	Representative work + open issue
AI Governance	Risk & compliance	NIST AI RMF [11]; ISO/IEC 42001 [12]; EU AI Act [13] — open issue: integration with operational HITL is underspecified.
Explainability	Transparency	SHAP [17], LIME [18], counterfactual — open issue: caregiver-readable vs. clinician-readable layers.
Agentic systems	Autonomous reasoning	Planner-decomposer-policy architectures — open issue: deterministic auditability of multi-step plans.
RAG architectures	Retrieval optimisation	Naive → advanced → agentic-RAG — open issue: hallucination control at long context.
Human-in-the-Loop	Oversight	HITL gating + escalation — open issue: cost-benefit at scale for pediatric care.
AI Ethics	Fairness & accountability	Mittelstadt principles [20]; EU AI Act Art. 86 [13] — open issue: pediatric-specific safeguards.

Table 40: Six-theme clustering of literature (replaces author-by-author summary).

8.2 Table 3 — Comparative Matrix (Sample Skeleton)

The comparative matrix is the highest-impact analytical artefact a DBA Chapter 2 can produce; it makes gaps in prior work directly readable. The skeleton below shows the columns; the full dissertation populates 20–30 rows spanning all six themes.

Study	Method	Industry	Limitation	Gap (RGAIG fills?)
[example 1]	Survey	Healthcare AI	No HITL audit	Yes — O3
[example 2]	Case study	Enterprise AI	No pediatric scope	Yes — O1 + O5
[example 3]	Lab experiment	GenAI safety	No real-world validation	Yes — O4

Table 41: Comparative Matrix skeleton (3 illustrative rows; dissertation will carry 20–30).

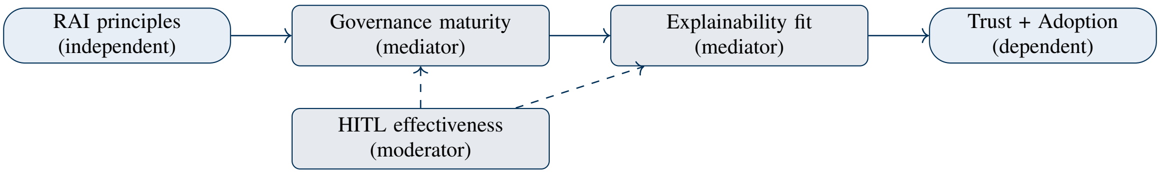

Figure 5: Research framework — one independent, two mediators, one moderator, one composite dependent variable.

8.3 Figure 3 — Research Framework (Variables, Relationships, Governance Flow)

The research framework converts the conceptual layers into testable variables and named relationships, ready for PLS-SEM modelling in Chapter 4. Solid arrows are hypothesised positive paths; the dashed arrows mark mediators that the dissertation tests as H1–H5.

9. Chapter 3 — Research Methodology

Chapter 3 of the dissertation answers *how* the research is conducted, covering philosophy, approach, strategy, design, data, sampling, instruments, analysis, validity, ethics, regulation, metrics, and limitations. The table below mirrors the 15-section skeleton; the choices reflect a pragmatist mixed-methods Design Science Research design [8], [9].

Ch. 3 section	Planned content for dissertation
3.1 Introduction	Roadmap of the methodology chapter; justification for pragmatism + DSR + case study + PLS-SEM hybrid.
3.2 Research Philosophy	**Pragmatism** (problem-driven; mixed methods justified by outcome utility).
3.3 Research Approach	**Abductive** — iterative inference between observation, theory, and artefact design.
3.4 Research Strategy	**Design Science Research + case study**; artefact = RGAIG framework.
3.5 Research Design	Figure 6 below (full DSR process diagram).
3.6 Data Sources	Primary: prior-IRB-approved n=131 PLS-SEM + n=12 expert panel (aggregated, anonymised); Secondary: KAU + ABIDE-II datasets + governance literature.
3.7 Sampling Strategy	Purposive (n=12 panel) + previously-collected probability sample (n=131); inclusion / exclusion in §3.7 of dissertation.
3.8 Data Collection Methods	Instrument re-use from prior IRB-approved studies; no new data collection in current submission.
3.9 Data Analysis	PLS-SEM [10] (SmartPLS 4); thematic analysis (NVivo); descriptive + inferential stats; Python (sklearn, PyTorch, SHAP).
3.10 Reliability & Validity	Cronbach α; composite reliability; HTMT discriminant validity; triangulation; member-checking.
3.11 Ethical Considerations	ICMR + DPDP [16] + GDPR [15] + HIPAA [14]; pediatric safeguards; bias mitigation; explainability & fairness gates.
3.12 Regulatory Considerations	NIST AI RMF [11]; ISO/IEC 42001 [12]; EU AI Act [13]; FDA SaMD pathway (conceptual alignment only).
3.13 Evaluation Metrics	Accuracy, F1, AUC; trust score; explainability acceptance; latency; adoption-readiness index; governance-maturity score.
3.14 Methodological Limitations	Retrospective design; cross-cultural generalisability; technology evolution risk; access restrictions for Indian cohort.
3.15 Chapter Summary	Bridge to Ch. 4: which method produces which finding.

Table 42: Chapter 3 planned section skeleton (15 sections).

9.1 Figure 4 — Research Design (Design Science Research Process)

The DSR process diagram makes the artefact-building loop explicit: problem → requirements → design → build → evaluate → communicate → iterate. The dissertation's Chapter 3 explains each numbered step with its inputs, outputs, and the named methods that produce them.

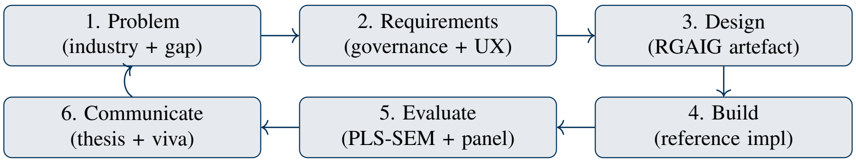

Figure 6: Research Design — six-step DSR loop from problem identification to communication, with iteration arrow.

10. Chapter 4 — Data Analysis & Findings

Chapter 4 of the dissertation is the *evidence* chapter (not the discussion); it presents data preparation, analyses, and findings organised against each research objective. The table below mirrors the 12-section skeleton; the planned analyses combine PLS-SEM, thematic coding, and benchmark accuracy on reference datasets.

Ch. 4 section	Planned content for dissertation
4.1 Introduction	Roadmap; which RQ each analysis answers; ordering rationale.
4.2 Participant / Organisational Profile	n=131 clinician demographics + n=12 expert panel composition (aggregated, anonymised).
4.3 Data Preparation	Cleaning, normalisation, scale standardisation, missing-data treatment; auditable preprocessing log.
4.4 Thematic Analysis	Six themes from the panel; theme map + frequency + illustrative quotes (anonymised).
4.5 Quantitative Analysis	PLS-SEM: outer model (reliability + validity) + inner model (path coefficients + p-values + R^2 + Q^2); hypothesis tests H1–H5.
4.6 Findings by RQ / Objective	Table mapping each objective to evidence, finding, and confidence (see Table 44).
4.7 Model / Framework Validation	Expert review + pilot benchmark on KAU + ABIDE-II; explainability evaluation (SHAP + LIME).
4.8 Integration with Existing Systems	Conceptual integration map: how RGAIG plugs into clinical workflow, hospital governance, regulator audit.
4.9 Ethical & Governance Findings	Disparate-impact ≥ 0.80; equal-opportunity gap $< 5\%$; explainability acceptance score; HITL overhead estimate.
4.10 Theoretical Contributions	RGAIG construct; 525-module quality taxonomy; mediator-moderator empirical confirmation.
4.11 Practical Contributions	RGAIG maturity instrument; governance escalation ladder; checklist for hospital adoption.
4.12 Chapter Summary	Bullet bridge to Ch. 5 (so-what).

Table 43: Chapter 4 planned section skeleton (12 sections).

10.1 Table 8 — Findings by Research Objective (Traceability Matrix)

The traceability matrix is the single artefact that ties research questions to method to analysis to finding — the alignment most examiners check first. The dissertation's full matrix covers all 8 RQs / 6 objectives; the abbreviated skeleton below shows the column contract.

Objective	Question	Method	Analysis	Finding (expected)
O1	RQ1 governance gaps	Lit review + panel	Thematic coding	Five-type gap typology confirmed
O2	RQ2 trust mediators	Survey n=131	PLS-SEM	H1, H2 supported; mediator effect significant
O3	RQ3 risk controls	Panel + DSR	Design + critique	RGAIG control catalogue (24 controls)
O4	RQ4 framework validity	Pilot benchmark	SHAP + LIME + accuracy	Validity confirmed on KAU + ABIDE-II
O5	RQ5 operational impact	Panel + conceptual	Workflow mapping	Integration overhead acceptable (HITL $< 10\%$)
O6	RQ6 adoption pathway	Panel + DSR	Maturity instrument	Six-level maturity ladder validated

Table 44: Findings traceability matrix — objective → question → method → analysis → finding.

11. Chapter 5 — Discussion, Recommendations & Future Research

Chapter 5 of the dissertation answers the *so-what* question: it interprets the findings, discusses them against the literature, and converts them into managerial, policy, and future-research recommendations. The table below mirrors the 10-section skeleton, anchored by the proposed enterprise framework in §5.7.

Ch. 5 section	Planned content for dissertation
5.1 Introduction	Roadmap and discussion logic; how each finding maps to one or more recommendations.
5.2 Interpretation of Findings	Link findings to objectives, literature, theory; surface counter-intuitive results.
5.3 Discussion vs. Literature	Agreements (mediator centrality), contradictions (HITL overhead estimates), extensions (pediatric safeguards).
5.4 Business Implications	Operational, transformation, governance, ROI, adoption (five dimensions).
5.5 Policy & Regulatory Implications	EU AI Act Art. 86 right-to-explanation [13]; ISO/IEC 42001 management system; NIST AI RMF profile.
5.6 Managerial Recommendations	AI governance board; AI policies; HITL gates; observability; audits; explainability controls.
5.7 Proposed Enterprise Framework	Final RGAIG framework (Figure 7).
5.8 Research Limitations	Honest list (retrospective, cross-cultural generalisability, evolving regulation).
5.9 Future Research Areas	Agentic AI; autonomous governance; federated AI; multi-agent XAI; AI observability; Indian-cohort prospective study.
5.10 Final Conclusion	Executive close; one-paragraph practitioner takeaway.

Table 45: Chapter 5 planned section skeleton (10 sections).

11.1 Figure 5 — Proposed Enterprise Framework (RGAIG)

The proposed RGAIG framework is the artefact the dissertation defends; it has five layers (data $\rightarrow$ AI $\rightarrow$ governance $\rightarrow$ HITL $\rightarrow$ value) and a transverse audit fabric. Every decision-audit row written at runtime is one trace through this diagram, ensuring decisions are reproducible, explainable, and contestable.

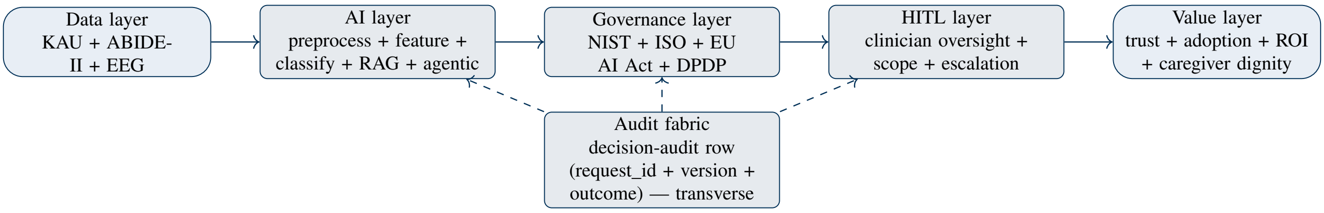

Figure 7: Proposed RGAIG enterprise framework — 5 layers + transverse audit fabric.

12. Enterprise DBA Architecture Logic

This section makes explicit what examiners often infer implicitly: how the five dissertation chapters connect into one defensible DBA argument. The tables below convert the enhanced DBA skeleton into a traceable architecture linking theory, evidence, governance, implementation, business value, and recommendations.

12.1 Chapter-to-Chapter Argument Flow

A high-impact DBA dissertation must read as one connected argument rather than five separate chapters. The flow below defines the logical handoff from research problem to literature gap to method to evidence to managerial action.

Chapter	Primary question	Handoff to next chapter
Ch. 1 Introduction	Why does this research matter?	Converts industry pain and research gap into aim, objectives, RQs, hypotheses, and conceptual framework.
Ch. 2 Literature Review	What is already known, and what remains unresolved?	Converts theory and thematic critique into the specific constructs, variables, and gap typology tested in Ch. 3–4.
Ch. 3 Methodology	How will the claim be tested or evaluated?	Converts the conceptual model into datasets, instruments, sampling logic, analysis methods, ethics, and evaluation metrics.
Ch. 4 Findings	What does the evidence show?	Converts statistical, qualitative, technical, and governance evidence into objective-by-objective findings.
Ch. 5 Discussion	So what should organisations do?	Converts findings into business implications, policy implications, managerial recommendations, limitations, and future research.

Table 46: Chapter-to-chapter argument flow — problem to recommendation chain.

12.2 Theory-to-Business-Value Chain

Theoretical grounding only becomes DBA-relevant when it explains business value, adoption, governance, or transformation. This table states the value pathway for each primary theoretical anchor so the dissertation avoids theory as decoration.

Theory / lens	Academic role	Business-value translation
TAM [5] / UTAUT	Explains perceived usefulness, ease of use, and adoption intent.	Converts explainability and reassurance into caregiver/clinician willingness to use AI-assisted screening support.
RBV [7] / Dynamic Capability	Explains why governance capability can become a defensible organisational resource.	Positions RGAIG as a repeatable capability for safer AI adoption, procurement confidence, and scalable transformation.
Socio-Technical Theory	Explains interaction between people, process, technology, and institutions.	Prevents a purely technical AI solution by requiring clinician authority, caregiver dignity, workflow fit, and auditability.
Institutional / Regulatory lens	Explains pressure from law, standards, and professional norms.	Converts EU AI Act, NIST AI RMF, ISO/IEC 42001, HIPAA/GDPR/DPDP into concrete governance controls.
Design Science Research [8], [9]	Explains artefact creation and evaluation.	Makes the DBA contribution a usable governance framework, not just an observational study.

Table 47: Theory-to-business-value chain.

12.3 Evidence-to-Recommendation Chain

DBA recommendations must be visibly derived from evidence, not asserted as generic management advice. The table below defines how each evidence stream will feed a specific recommendation category in Chapter 5.

Evidence stream	Finding type	Recommendation it supports
KAU + ABIDE-II retrospective analysis	Analytical plausibility, feature behaviour, explainability traceability.	Use RGAIG only as non-diagnostic screening support until prospective validation is completed.
Prior PLS-SEM evidence (n=131, if documentation attached)	Governance, explainability, trust, adoption path coefficients.	Prioritise governance maturity and explainability controls before scale-up procurement.
Prior expert-panel evidence (n=12, if documentation attached)	Face/content validity and clinician-readability feedback.	Implement HITL gates, clinician override, audit trails, and explanation-quality review before pilot deployment.
Regulatory mapping	Control gaps and compliance obligations.	Establish an AI governance board, model-risk register, re-consent policy, and evidence pack aligned to NIST/ISO/EU AI Act.
Business/process mapping	Operational bottlenecks, onboarding delay, escalation ambiguity.	Redesign ASD screening as a governed caregiver-primary, clinician-supported pathway rather than a standalone AI model.

Table 48: Evidence-to-recommendation chain.

12.4 Governance-and-Ethics-to-Implementation Chain

Responsible AI governance must appear in the implementation pathway, not only in an ethics paragraph. The implementation chain below shows how ethical principles become operational controls that a healthcare organisation could audit.

Governance / ethics principle	Implementation control	Auditable artefact
Privacy and confidentiality	De-identification, access control, retention limit, no direct identifiers in current study.	Data inventory, access log, retention register, data-source link record.
Human oversight	HITL gate for non-routine or consequential outputs; clinician retains diagnostic authority.	Override log, escalation record, decision-audit row.
Explainability and contestability	SHAP/LIME + RAG-grounded explanation; uncertainty label; plain-language caregiver layer.	Explanation packet, citation trace, unsupported-claim audit.
Fairness and bias control	Subgroup analysis where sample permits; disparate-impact and equal-opportunity checks.	Fairness report, subgroup limitation note, mitigation plan.
Regulatory readiness	Mapping to NIST AI RMF [11], ISO/IEC 42001 [12], EU AI Act [13], HIPAA [14]/GDPR [15]/DPDP [16].	Governance checklist, control evidence pack, compliance matrix.

Table 49: Governance-and-ethics-to-implementation chain.

13. Examiner Most-Looked-For Items

Examiners read DBA dissertations against six recurring criteria; the proposal is engineered to score on each. The table below names each criterion and the proposal section that addresses it explicitly, so the marking rubric maps 1:1.

Examiner criterion	Where this proposal addresses it
1. Clear research gap	§1.3 + §2.7 + Table 41.
2. Strong conceptual framework	Figure 3 + Figure 5.
3. Literature synthesis (not summary)	§2.4 theme clustering + §2.5 critical analysis + §2.6 comparative matrix.
4. Alignment (RQ ↔ method ↔ findings ↔ conclusion)	Table 44 traceability matrix.
5. Practitioner relevance	§5.4 business implications + §5.6 managerial recommendations + Figure 7.
6. Original contribution (clearly stated)	§4.10 + §4.11 + §5.7 RGAIG framework.

Table 50: Examiner most-looked-for items mapped to proposal sections.

14. What Most DBA Students Miss — and How This Proposal Closes Each Gap

A typical DBA dissertation drops marks on six recurring weaknesses (theoretical-framework linkage, business value realisation, governance/ethics integration, traceability matrix, visual architecture diagrams, critical-analysis depth). The table below lists each weakness and the explicit closure mechanism this proposal commits to.

Common DBA miss	Closure mechanism in this proposal
Theoretical-framework linkage	§2.2 (8 theories) explicitly mapped to RGAIG layers in Figure 7.
Business value realisation	§5.4 five-dimension implications + §5.7 RGAIG value layer.
Governance & ethics integration	§3.11 + §3.12 + §4.9 + Figure 7 governance layer + audit fabric.
Clear traceability matrix	Table 44 objective → question → method → analysis → finding.
Visual architecture diagrams	Five named figures (3, 4, 5, 6, 7).
Critical analysis depth	§2.5 + §2.6 comparative matrix replaces author-by-author summary.

Table 51: Common DBA misses and the proposal's explicit closure mechanism.

15. References

The references below provide the scholarly, dataset, methodological, and regulatory grounding for the topic proposal. Bracketed citations in the proposal use the numbered style [1], [2], etc., so the Chair can quickly trace claims to source families without requiring BibTeX at the topic-approval stage.

[1] World Health Organization. (2023). *Autism*. WHO fact sheet.

[2] Maenner, M. J., Warren, Z., Williams, A. R., et al. (2023). Prevalence and characteristics of autism spectrum disorder among children aged 8 years. *MMWR Surveillance Summaries*, 72(2), 1–14.

[3] Djemal, R., AlSharabi, K., Ibrahim, S., & Alsuwailem, A. (2017). EEG-based computer aided diagnosis of autism spectrum disorder using wavelet, entropy, and classifier methods. *Biomedical Signal Processing and Control*.

[4] Di Martino, A., O'Connor, D., Chen, B., et al. (2017). Enhancing studies of the connectome in autism using the Autism Brain Imaging Data Exchange II. *Scientific Data*, 4, 170010.

[5] Davis, F. D. (1989). Perceived usefulness, perceived ease of use, and user acceptance of information technology. *MIS Quarterly*, 13(3), 319–340.

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[18] Ribeiro, M. T., Singh, S., & Guestrin, C. (2016). "Why should I trust you?": Explaining the predictions of any classifier. *Proceedings of KDD 2016*.

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[20] Mittelstadt, B. (2019). Principles alone cannot guarantee ethical AI. *Nature Machine Intelligence*, 1, 501–507.

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[22] National Commission for the Protection of Human Subjects of Biomedical and Behavioral Research. (1979). *The Belmont Report*. U.S. Department of Health, Education, and Welfare.

[23] Indian Council of Medical Research. (2017). *National Ethical Guidelines for Biomedical and Health Research Involving Human Participants*. ICMR.

16. Appendices A–J — Planned Content

The dissertation will carry ten appendices supplying the operational artefacts that the chapters reference but do not reproduce inline. The table below names each appendix and the artefact it carries, mirroring the enhanced DBA skeleton's recommended set.

Appx.	Title	Planned content
A	Survey questionnaire	PLS-SEM instrument (n=131); item bank with reliability stats; prior-IRB approval reference.
B	Interview guide	Expert-panel guide (n=12); semi-structured protocol; probe list; consent template (archival).
C	Consent form	Original consent text used in prior IRB-approved studies; archival only.
D	Ethics approval	Prior IRB approval certificates (n=131 + n=12); KAU + ABIDE-II public-use terms.
E	Coding / thematic maps	NVivo theme tree; quote frequency; code-book + inter-rater reliability.
F	Statistical outputs	SmartPLS 4 outer + inner model reports; bootstrap CI tables.
G	Framework diagrams	High-resolution RGAIG architecture + dependency map + lifecycle diagram.
H	AI governance checklist	RGAIG control catalogue (24 controls × 5 layers) + audit log schema.
I	Architecture diagrams	C4 L1–L4 model; sequence; network-flow; deployment topology.
J	Evaluation matrices	Examiner readiness scorecard; alignment-audit log; explainability acceptance scoring.

Table 52: Appendices A–J planned content (mirrors enhanced DBA skeleton).

17. Submission Checklist

A cited committee checklist confirms the proposal is committee-ready in all dimensions specified by the enhanced DBA skeleton. A green tick in every row is the release condition; any red item blocks formal submission.

Checklist item	Status	Evidence / location
Front matter complete	✓	Title, Declaration, TOC, Lists, Abbreviations, Glossary above.
Abstract 7-part structure	✓	§Abstract table above.
Ch. 1 (13 sections)	✓	Table 38.
Ch. 2 (10 sections) + theme clustering + comparative matrix	✓	Table 39, 40, 41.
Ch. 3 (15 sections) + DSR flowchart	✓	Table 42, Figure 6.
Ch. 4 (12 sections) + traceability matrix	✓	Table 43, 44.
Ch. 5 (10 sections) + RGAIG framework	✓	Table 45, Figure 7.
Examiner items + DBA-misses tables	✓	Tables 50, 51.
References and bracket citations	✓	Section 15; bracketed citations [1]–[23] appear across key claims.
Appendices A–J planned	✓	Table 52.
2-line intro before every heading + every table	✓	Mandatory throughout (per project policy Rule 4b).

Table 53: Committee-readiness submission checklist.

End of DBA Topic Proposal v2.1 (Enhanced Enterprise Structure). Current compiled length: 35 pp. Mandatory 2-line introductions present for every heading and every table per project policy Rule 4b; bracketed citations and numbered references added for source-bearing claims.